

Java Unit - 4

Section - A

Q1) What are byte streams in Java?

Ans) Byte streams are used to handle binary data such as images, audio, video or any other non-textual data. They read or write data in the form of individual bytes.

Some classes for byte stream handling in Java are:
InputStream and OutputStream.
FileInputStream and FileOutputStream.
BufferedInputStream and BufferedOutputStream.

Q2) Define character streams in Java.

Ans) Character streams are used to handle textual data, such as reading or writing characters or strings. They read or write data in the form of characters.

Some classes for character streams handling in Java are:
Reader and Writer.
FileReader and FileWriter.
BufferedReader and BufferedWriter.
InputStreamReader and OutputStreamWriter.

Q3) What is the purpose of the File class in Java?

Ans) This class represents a file or directory path in the file system. It provides methods to create, delete and check the existence of files and directories.

Q6) Explain the use of the FileInputStream in Java.
Ans) FileInputStream is a class that represents an input stream of bytes. It's used to read data from a file. This data could be anything from text to binary data.

Q5) Define FileOutputStream, and how is it used?
Ans) FileOutputStream is a class that represents an output stream of bytes. It's used to write data from a file. This data could be anything from text to binary data.

Q6) Define InputStreamReader class in Java?

Ans) InputStreamReader class is used to convert byte streams to character streams. It reads bytes from an InputStream and decodes them into characters according to a specified character encoding.

Q7) Explain the purpose of OutputStreamWriter in Java?

Ans) OutputStreamWriter class is used to convert character streams to byte streams. It provides character encoding and decoding capabilities.

Q8) Write the differences b/w Reader and Writer classes in Java?

Ans) The Reader and Writer are the abstract classes for all character input and output streams, respectively. They provide methods for reading and writing characters.

Q9) Explain the purpose of `BufferedReader` class.

Ans) A `BufferedReader` in Java is a class designed to read text from a character input stream buffering characters to provide for the efficient reading of characters. They improve performance by reducing the number of I/O operations.

Q10) Define the `List` class in the Java Collection Framework?

Ans) The class `List` in the Java Collection Framework represents an ordered collection of elements. It allows duplicate elements and maintains the insertion order of elements.

There are three `List`:

- 1) `ArrayList`
- 2) `LinkedList`
- 3) `Vector`

Q11) What is the difference between `ArrayList` and `LinkedList`?

Q12) What are the rules of ArrayList class?
Ans) Rule 1) :- The class ArrayList implements from List interface and List extends collection framework.

Rule 2) :- Class ArrayList is present in java.util Pack.

Rule 3) :- Syntax of ArrayList :-

class Test

{

public static void main (String [] args)

{

ArrayList al = new ArrayList ();

//method

}

}

Rule 4) :- ArrayList created on the basis of Resizable Array.

Q13) Explain the difference b/w Doubly and circular linked list? Make a Diagram node only.

Java Unit - 2

Q1) Define a class and an object in Java?

- Ans:-
- class is a collection of objects.
 - class is not a real world entity. It is just a Template or blueprint.
 - class does not occupy memory.

Syntax:-

```
Access modifiers class ClassName  
{  
    method  
    constructors  
    variable  
    blocks  
    Nested class  
}
```

Object :-

- object is an instance of class.

- object is a real world entity.
- object occupies memory.

Syntax:-

```
int a = new int ( );
```

Q2) What is the purpose of the new operator in Java?

Ans:-